Streaming Data Analytics

Our Research Activities and Thesis Offering

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Politecnico di Milano

POLITECNICO
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How to apply for a thesis with us

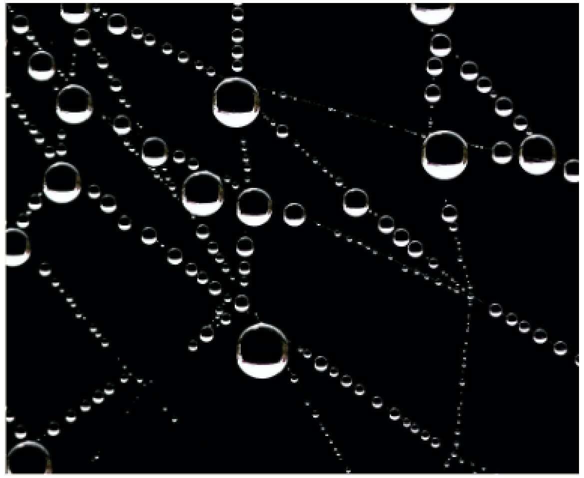
If you are interested in one of the illustrated topics

- Fill out this form: https://forms.office.com/e/E8d9Da314h
- Or scan the Qrcode:

Stream Reasoning

Where it all begun, 2008-2017

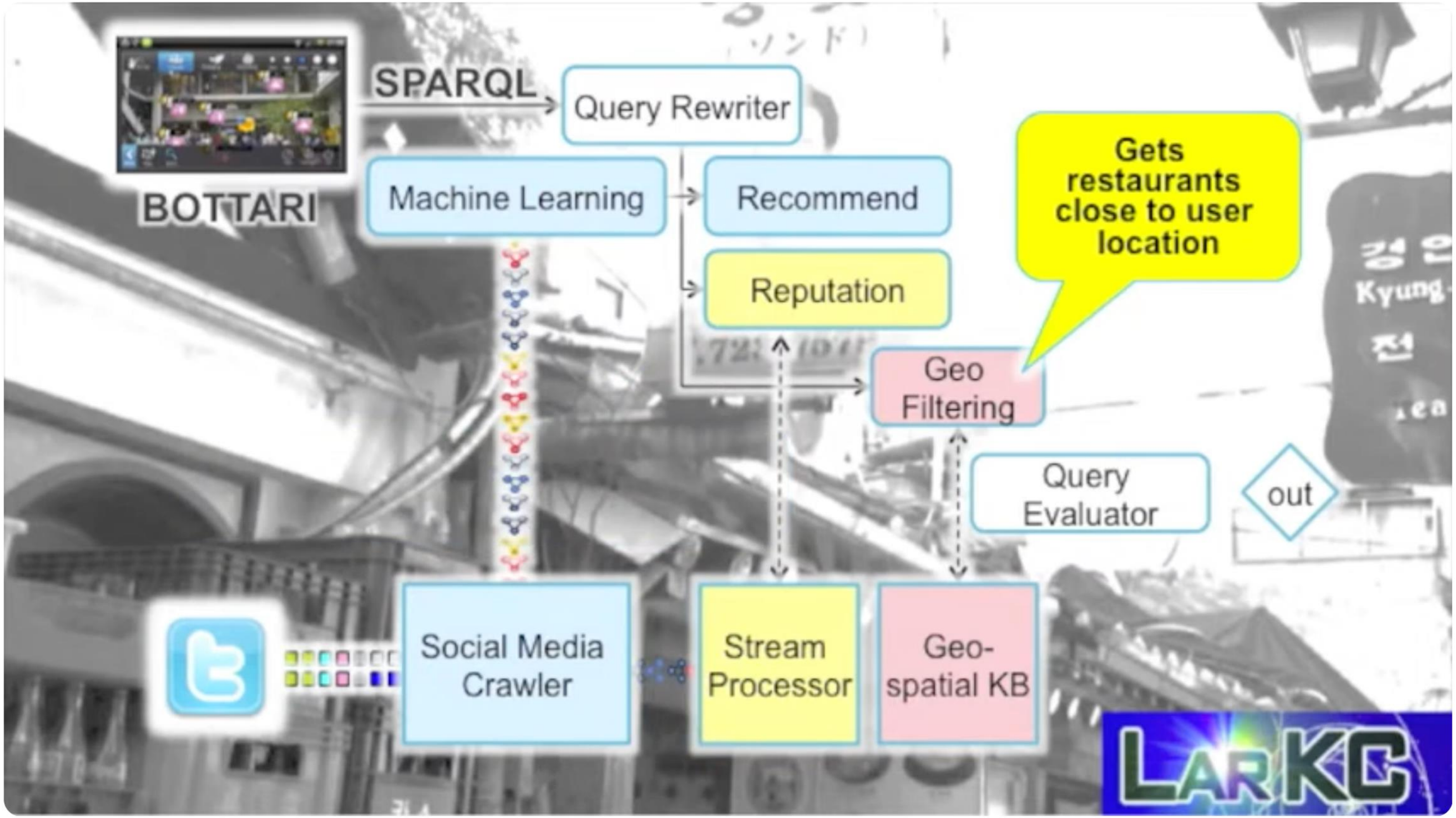
https://web.archive.org/web/20111128223400/http://streamreasoning.org/events/sr2009



It's a Streaming World! Reasoning upon Rapidly Changing Information

Emanuele Della Valle and Stefano Ceri, *Politecnico di Milano*
Frank van Harmelen, *Vrije Universiteit Amsterdam*
Dieter Fensel, *University of Innsbruck*

https://doi.org/10.1109/MIS.2009.125



https://youtu.be/XGOKe_lhSks

Big Stream Processing Systems
(Oct 29 – Nov 03, 2017)

https://www.dagstuhl.de/seminars/seminar-calendar/seminar-details/17441

Our Current Research

Blending Time Series Analytics, Streaming Machine Learning & Continual Learning

Data Science
Volume 6, Issue 1-2, 2023, Pages 1-16

https://doi.org/10.3233/DS-220057

Research Article

Towards time-evolving analytics: Online learning for time-dependent evolving data streams

Giacomo Ziffer [a,*], Alessio Bernardo [b], Emanuele Della Valle [c], Vitor Cerqueira [d], and Albert Bifet [e]

Abstract
Traditional historical data analytics is at risk in a world where volatility, uncertainty, complexity, and ambiguity are the new normal. While Streaming Machine Learning (SML) and Time-series Analytics (TSA) attack some aspects of the problem, we still need a comprehensive solution. SML trains models using fewer data and in a continuous/adaptive way relaxing the assumption that data points are identically distributed. TSA considers temporal dependence among data points, but it assumes identical distribution. Every Data Scientist fights this battle with ad-hoc solutions. In this paper, we claim that, due to the temporal dependence on the data, the existing solutions do not represent robust solutions to efficiently and automatically keep models relevant even when changes occur, and real-time processing is a must. We propose a novel and solid scientific foundation for Time-Evolving Analytics from this perspective. Such a framework aims to develop the logical, methodological, and algorithmic foundations for fast, scalable, and resilient analytics.

https://doi.org/10.3233/ds-220057

EDITOR: San Murugesan, san@computer.org

COLUMN: AI FOCUS

Streaming Continual Learning for Unified Adaptive Intelligence in Dynamic Environments

Federico Giannini and Giacomo Ziffer, *Dipartimento di Elettronica, Informazione e Bioingegneria, Politecnico di Milano, Milano, 20133, Italy*

Andrea Cossu and Vincenzo Lomonaco, *Department of Computer Science, University of Pisa, Pisa, 56127, Italy*

https://doi.org/10.1109/MIS.2024.3479469

Understanding our Thesis

A schema to frame the six types of thesis we offer

Technology Readiness Levels

TRL 9	System proven in operational environment
TRL 8	System complete and qualified
TRL 7	**Integrated pilot system demonstrated**
TRL 6	**Prototype system verified**
TRL 5	**Laboratory testing of integrated system**
TRL 4	**Laboratory testing of prototype component or process**
TRL 3	**Critical function, proof of concept established**
TRL 2	Technology concept and/or application formulated
TRL 1	Basic principles are observed and reported

industrial

Also, in collaboration with our spin-off (Motus ml) and its customers (e.g., Thales Alenia Space, Leonardo Space, etc.)

research

Also, in collaboration with foreign universities, e.g., INSA Lyon

State of the Art vs. State of the Practice

State of the Art

- the **highest level of development of a** technique or **scientific field**, achieved at a particular time
- In our context, it refers to the latest and most advanced streaming data engineering and streaming data science **academic results**

State of the Practice

- the current accepted methodologies, techniques, tools, and approaches that are **widely used in a particular industry**
- In our context, it involves using widely accepted systems, tools, and methodologies that are **commonly used by Data Engineers and Data Scientists** in their day-to-day processing of data streams

Theses's types

	Novel result	Benchmarking	Use case
Research	Develop a novel algorithm/method so that the result can be the content of a **scientific publication** **TRL 3**	**Compare** multiple **state-of-the-art methods** in several previously published settings to determine the **trade-offs** **TRL 4**	Investigate which is the best way to address a real-world problem using **state-of-the-art** solutions **TRL 5**
Industrial	Develop a novel algorithm/component so that the result can be the core of a **new product/service** **TRL 5**	**Compare** multiple **state-of-the-practice solutions** in several real-world conditions to determine the **trade-offs** **TRL 6**	Investigate which is the best way to **address a real-world problem** using **state-of-the-practice** solutions **TRL 7**

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E.g., S. D'Andrea: **Growing intelligent networks for streaming continual learning with temporal dependence**

https://www.politesi.polimi.it/handle/10589/234823

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E.g., M.Allam: [...] **Optimizing streamDM-C++ Library for Efficient Streaming ML Algorithms on Microcontrollers**

https://www.politesi.polimi.it/handle/10589/214278

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E.g., S. Menconi: **Comparing traditional and Streaming Machine Learning methods for soccer pass detection**

https://www.politesi.polimi.it/handle/10589/210996

Theses's types

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E.g., A. Persici: [...] **harnessing** [...] **distributed computing for advanced analytical** [...] **astronomical datasets**

https://www.politesi.polimi.it/handle/10589/231560

Theses's types

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E.g., N. Francescon: [...] **Streaming Machine Learning for real-time land use classification in satellite imagery**

https://www.politesi.polimi.it/handle/10589/230669

Theses's types

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E.g., F. Renzi: **Online learning techniques for occupancy detection on resource constrained devices**

https://www.politesi.polimi.it/handle/10589/218745

Streaming Continual Learning

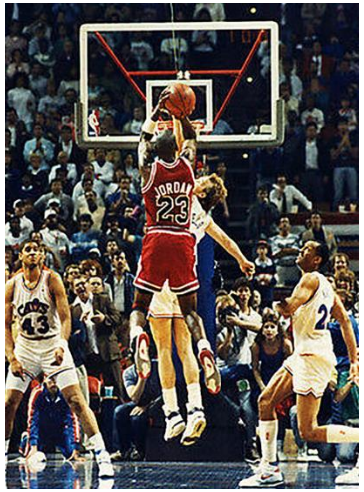
Federico Giannini

Streaming Continual Learning

	Streaming ML	Continual Learning	Time Series Analysis	SCL
Drift detection.	✓			✓
Quick adaptation.	✓	Only input drifts		✓
Learning continuously.	✓	Online CL		✓
Avoiding forgetting.		✓		✓
Temporal dependence.			✓	✓

Streaming Continual Learning

SML



CL

TSA

Streaming Continual Learning

What does it mean to avoid forgetting when concepts may be in contradiction?

- Maintain a **complete and general knowledge** of the entire data stream.
- **DON'T LOOK BACK IN ANGER!**

  Look back at the previous knowledge and select/modulate the helpful portions that can be reused. Not all the knowledge is invalidated, different concepts may share similarities over time.

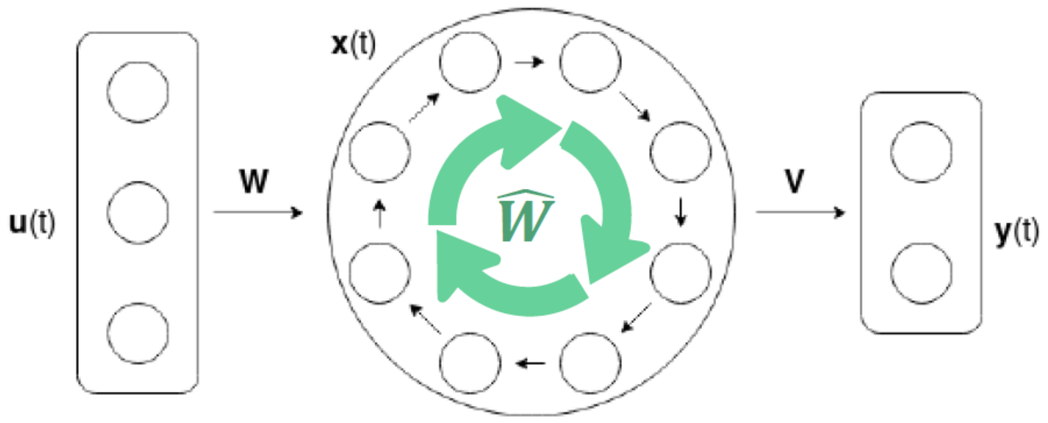
Research Novel Result Thesis

Echo State Networks for SCL (In collaboration with Università di Pisa)

Echo State Networks (ESNs) are a class of recurrent neural networks that are particularly well-suited for modeling temporal dynamics, thanks to their fixed recurrent reservoir and efficient training of the readout layer.

The objective of this thesis is to apply ESNs in the context of SCL to effectively capture temporal dependencies among continuously arriving data points.

For details, see: ***"26c-Echo_State_Network_for_SCL_with_Unipi.pdf"***.



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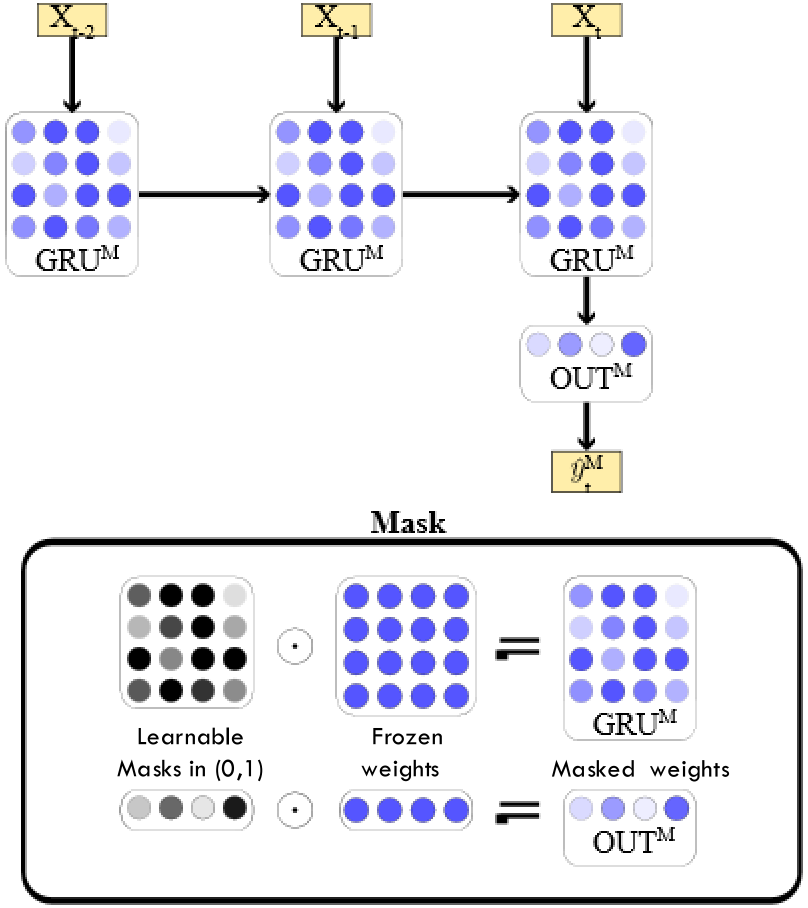
claudio.gallicchio@unipi.it

Research Novel Result Thesis

CL Architectural Strategies for SCL with Temporal Dependence (Deep Learning background required)

CL Architectural strategies can freeze the weights, isolate portions of the network, use transfer learning, and expand the architecture. These abilities allow the solution to focus on one concept at a time, reuse previous knowledge, and build a cumulative general knowledge (i.e., avoid forgetting). The goal of the thesis is to select a CL strategy to be applied on top of a continuous recurrent neural network to embody the SCL vision.



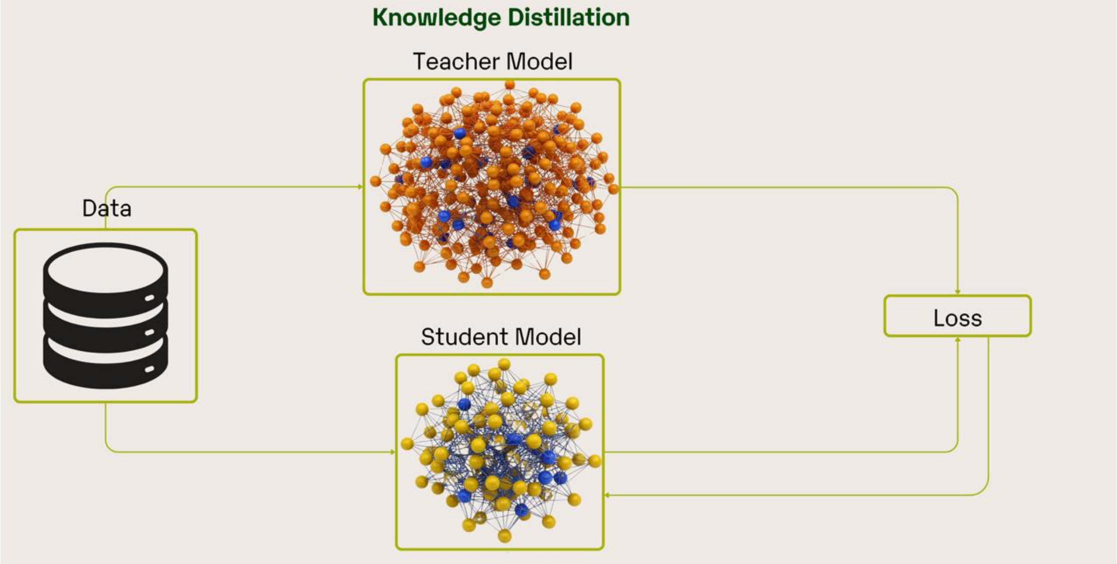
Research Novel Result Thesis

Continuous Distillation for SCL (Deep Learning background required)

When applying architectural strategies, the dimension of the model grows with the number of drifts. This thesis proposes the study of a continuous distillation strategy to control model complexity over time.

The goal is to design methods that identify and remove unused or redundant parts of the architecture, compact neurons or layers during continuous learning, or build a smaller model that mimics the behavior of the original one through Knowledge Distillation.

Such an approach aims to maintain adaptability to concept drift while keeping the model efficient and scalable.



Research Novel Result Thesis

Stable Hoeffding Adaptive Trees for SCL

HATs focus on the current concept to provide highly adaptive solutions. This strong adaptability may lead to *catastrophic forgetting* when parts of the decision tree are pruned after a concept drift.

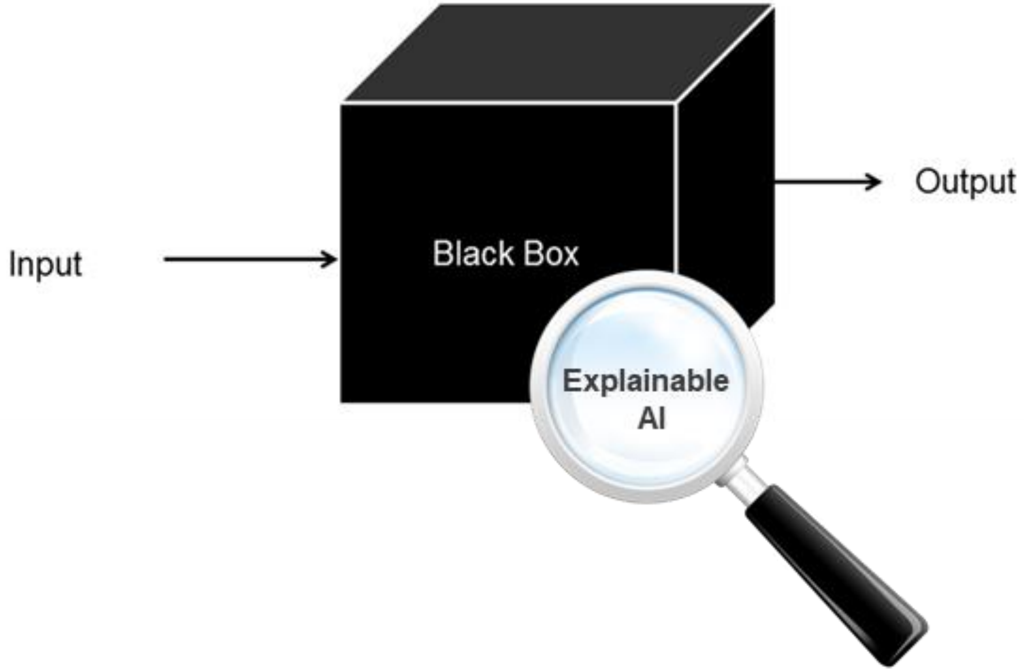
This thesis investigates strategies to mitigate forgetting by storing the pruned subtrees and reintroducing them after subsequent drift detections, when previously learned concepts may become relevant again. In addition, the work aims to analyze which portions of the tree are not affected by the drift and therefore should not be pruned, improving knowledge retention while maintaining the adaptive nature of the model.

Research Novel Result Thesis

Continuous Explainability

Understanding why SCL models work and which patterns they recognize is crucial for building trustworthy and reliable systems. In streaming scenarios, models continuously adapt to evolving data distributions, making their behavior dynamic and harder to interpret compared to static models.

Providing explainability in this context requires methods that can generate explanations incrementally, track how decision patterns change over time, and relate model updates to detected concept drifts. This thesis explores approaches for *streaming explainability*, aiming to explain predictions, model updates, and learned patterns in real time, while accounting for the non-stationary nature of the data and the continuous evolution of the model.



Research Novel Result Thesis

Hyper Parameter Tuning for SCL (Deep Learning background required)

A major challenge in streaming solutions is the selection of model hyperparameters, which are typically set empirically. In a non-stationary setting, hyperparameters strongly influence the performance, yet their optimal configuration may change over time.

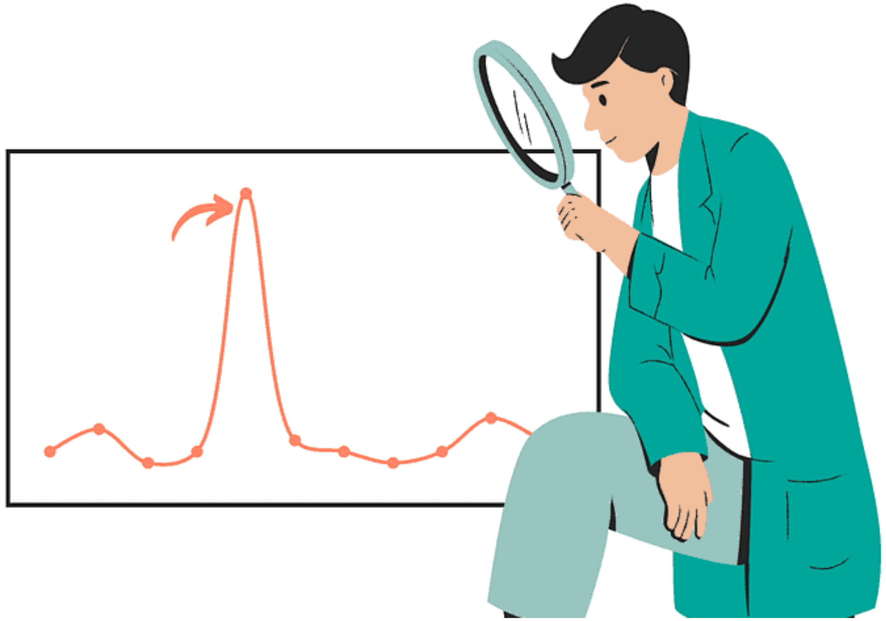
Despite their importance, current literature does not provide a definitive solution for principled hyperparameter tuning in streaming scenarios. This thesis addresses this gap by investigating strategies for adaptive hyperparameter selection.

Research Novel Result Thesis

SCL Anomaly Detection (Deep Learning background required)

SML and CL are typically studied in the context of classification tasks. However, extending this framework to anomaly detection represents an important and promising direction. In many real-world scenarios, anomalies must be detected without access to labeled data, making unsupervised approaches essential.

This thesis explores the extension of SCL methods to anomaly detection settings. By avoiding the need for labeled data, this approach increases the applicability of streaming continuous learning to realistic and large-scale environments.



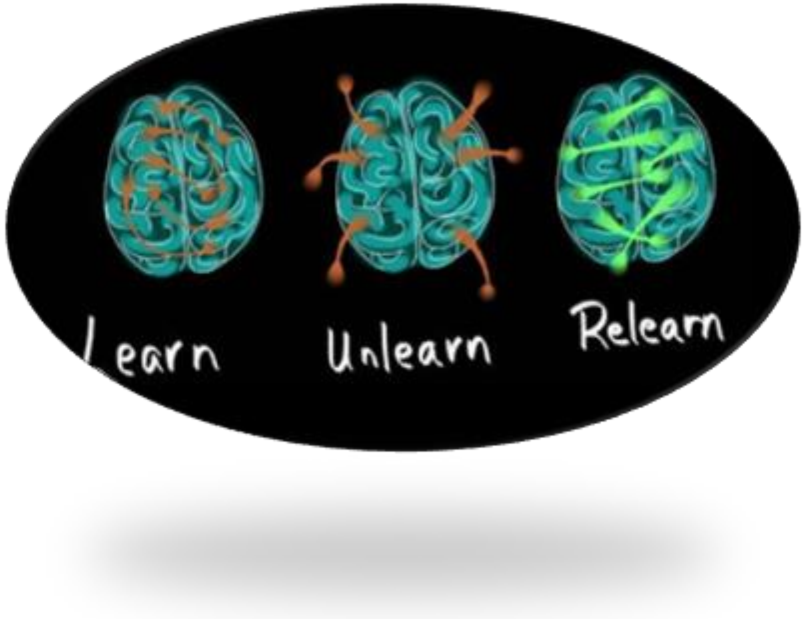
Research Novel Result Thesis

Machine Unlearning for SCL (Deep Learning required)

In the presence of a contradictory drift, previously learned knowledge becomes invalid because the underlying problem itself changes. In such cases, the model is required to forget part of what it has learned in the past in order to remain correct and effective.

A key open challenge is understanding what knowledge should be forgotten and when. How can we identify the components of the model that are no longer valid after a contradictory drift? How can we selectively and efficiently remove or overwrite this knowledge without harming still-relevant information?

This thesis will investigate mechanisms for detecting obsolete knowledge and guiding controlled forgetting in streaming learning models, leveraging concepts from machine unlearning to remove outdated information efficiently while preserving useful knowledge.

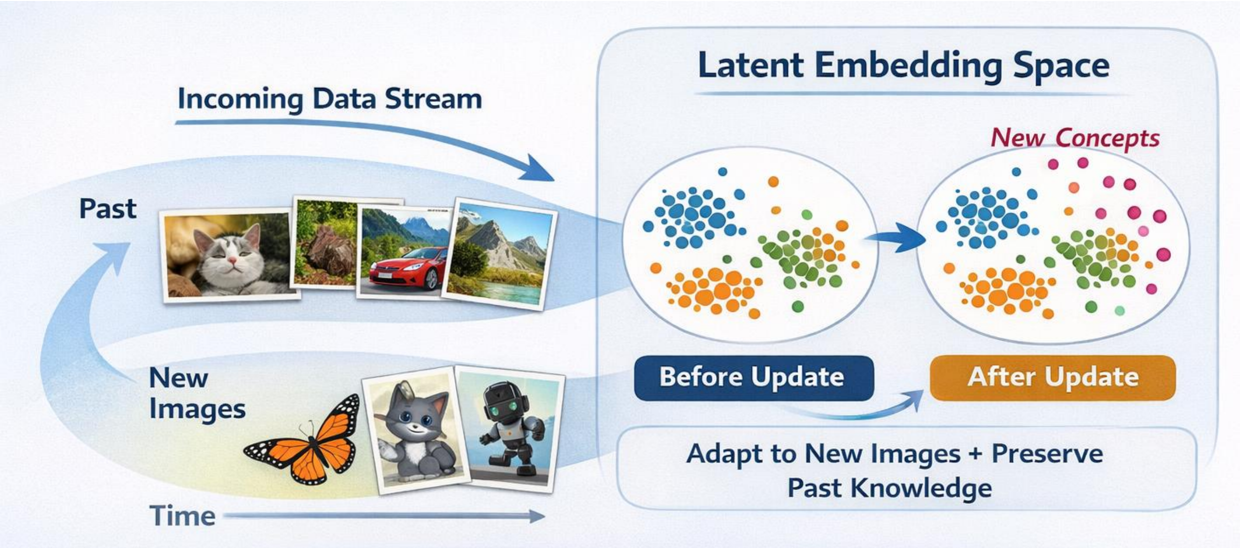


Research Novel Result Thesis

Continuous Self Supervised Learning (Deep Learning background required)

Continuous learning can also be applied to self-supervised pre-training of embeddings. In streaming scenarios, it is necessary to continuously learn image representations without forgetting how to represent previously observed images. At each update, the latent embedding space must remain consistent with the previous one, so that past images are still correctly represented.

Since data arrive continuously over time, the model is progressively exposed to new types of images. This thesis addresses this problem by investigating methods that ensure stability and consistency of the latent representation space while allowing the model to adapt to new visual concepts in a continuous self-supervised setting.



Continuous Multimodal Foundation Models

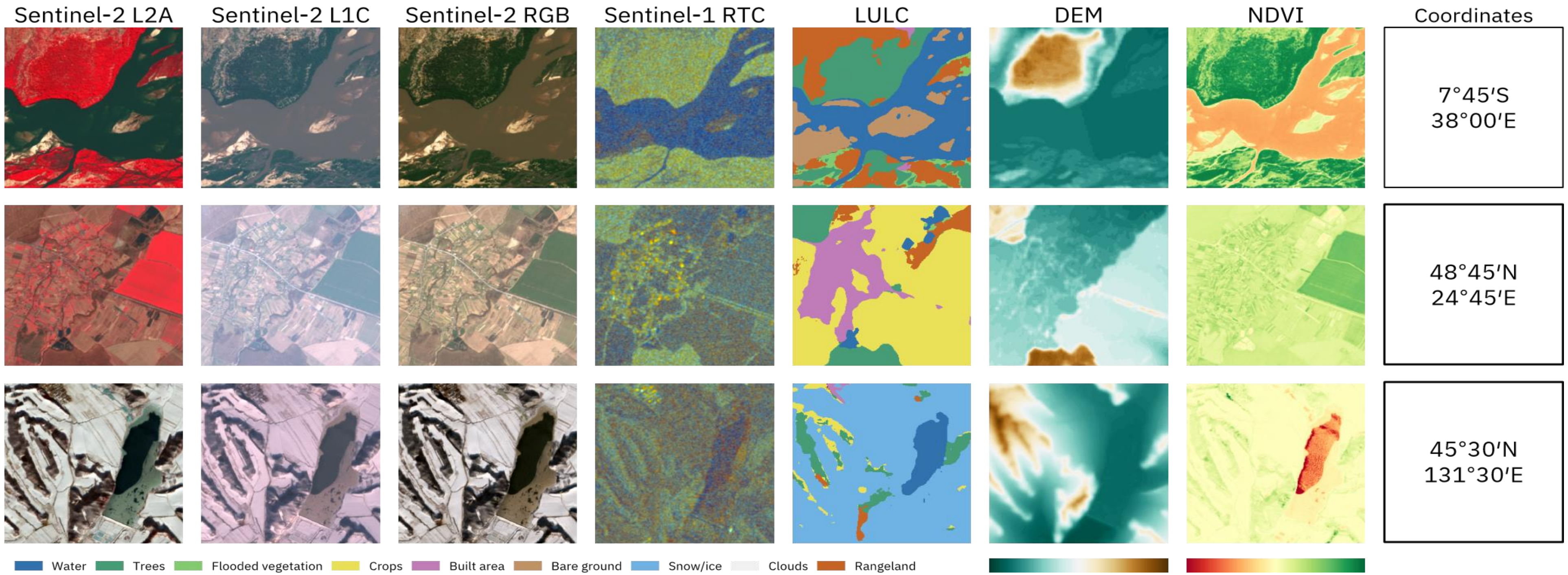
Marcello Declich, Federico Giannini

What is a Foundation Model?

What is a Foundation Model?

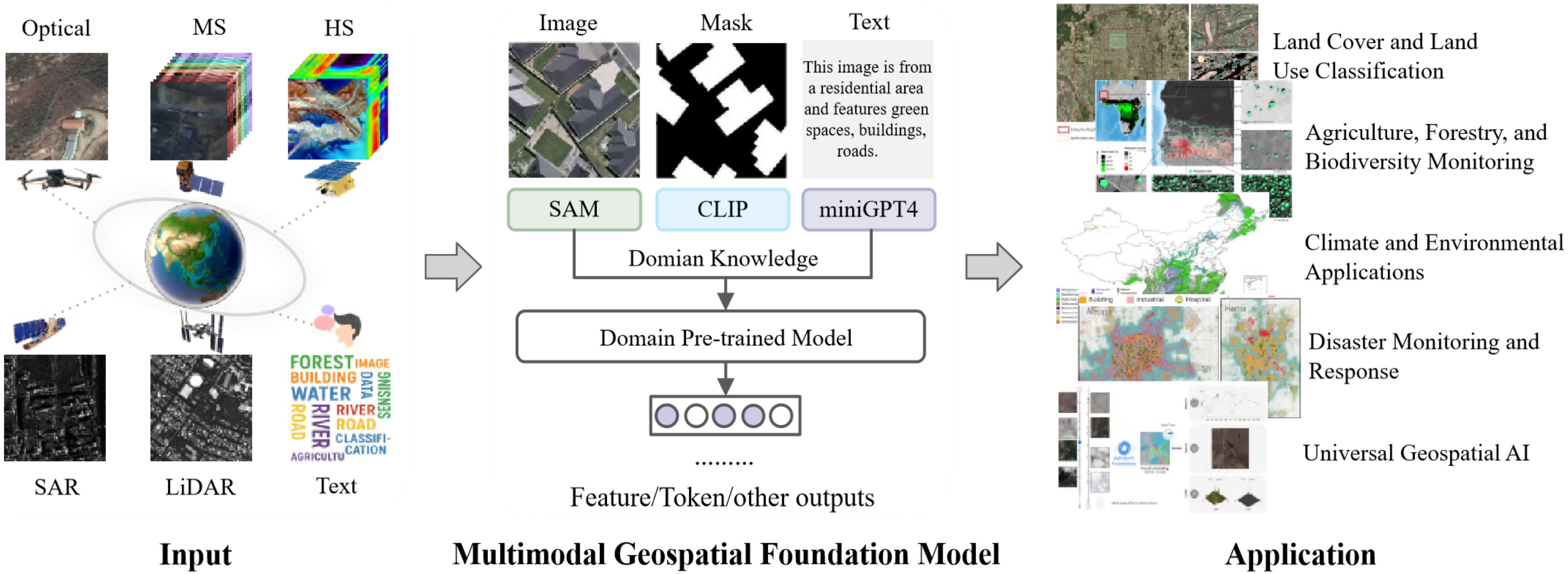
- LLMs
- Text-to-video
- Text-to-photo
- CLIP, SAM, etc…

Geospatial Foundation Model



Jakubik et al. 2025

Geospatial Foundation Model



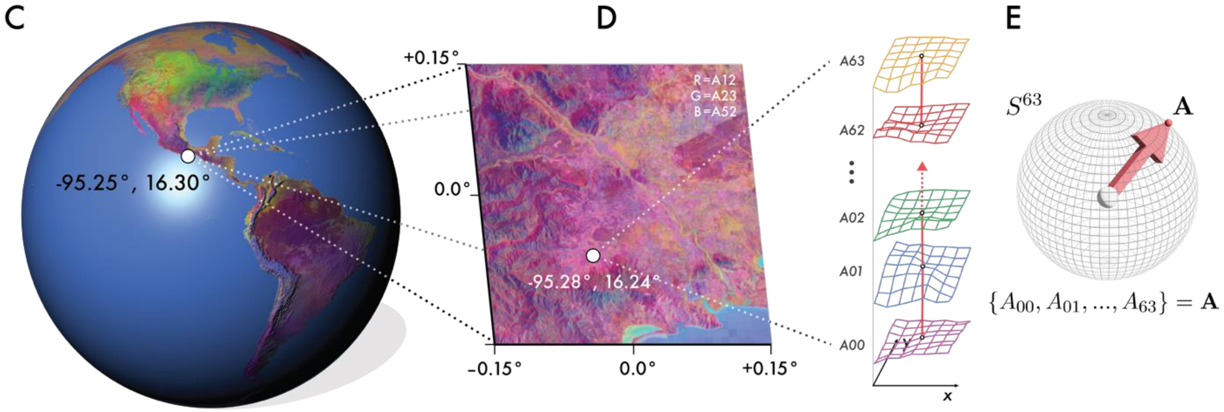
Geospatial Foundation Model

Jakubik et al. 2025

Think about how many applications could be developed if one of these models were placed on board a satellite?

1. Satellite data will be used the moment it's captured. No waste of time for downlink. Think about all the time-critical applications that can benefit.

2. Missions become much more flexible. Model components can be updated on the fly during a satellite's lifetime.



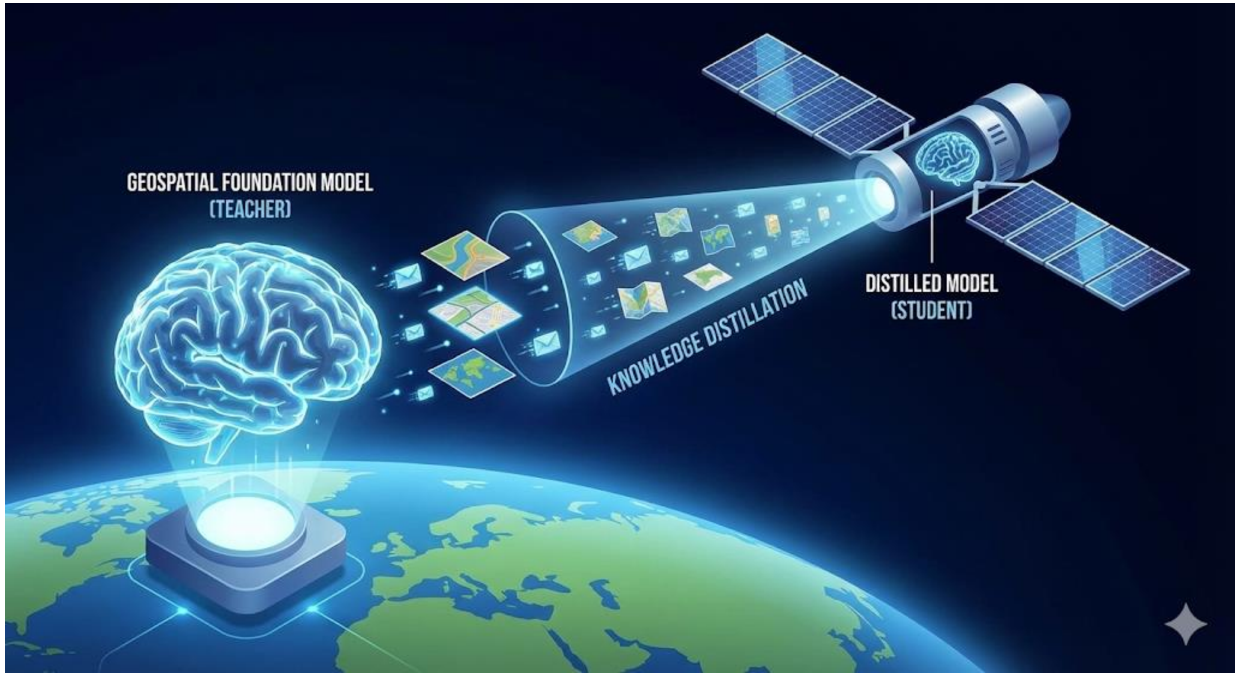
Research/Industrial Novel Research Thesis

Knowledge Distillation of Geospatial Foundation Models

Goal

The release of Geospatial Foundation Models is revolutionizing Earth Observation. However, their computational cost makes them unsuitable for real-time applications on edge devices, such as drones or satellites.

This thesis aims to bridge the gap between high-performance AI and edge deployment. The goal is to research and implement Knowledge Distillation from a large Geospatial Foundation Model into a compact lightweight network, focusing on the **trade-off** between accuracy and inference speed, with a particular focus on **cross-modal distillation.**




WHEN WILL ONE OF THESE MODELS BE ABLE TO WORK IN SPACE?
A. 1 YEAR
B. 2 YEARS
C. 5 YEARS
D. 10 YEARS

Johannes Jakubik • Già segui
Research Scientist @ IBM Research | PhD in Deep Learning
2s •

The most remote version of our TerraMind model is now orbiting Earth at 520km altitude -- farther away from us than the ISS 🛰️

Big moment for our team in IBM Research Europe following the launch of a SpaceX Falcon-9 with payload carrying TerraMind.tiny, our latest multimodal generative deep learning model, into orbit. Huge kudos to Planetek Italia and European Space Agency - ESA for making this possible!

Why is this exciting? A few years ago, running a 5 million parameter multimodal generative model in real-time in space would have sounded like science fiction. Today, it does not anymore. 😉 To me, this shows how fast innovation cycles between hardware, software, and modeling have become. It will be a long road until we see AI in space at scale, but this mission already is a fantastic interdisciplinary achievement.

- [TerraMind tiny linkedin post]

There are already models in Space above our heads!!

Think about the implications for the future:
- Satellite data will be used the moment it's captured. No waste of time for downlink. Think about all the time-critical applications that can benefit.
- Missions become much more flexible. Model components can be updated on the fly during a satellite's lifetime.
- Researchers and practitioners can download our models from Hugging Face knowing that they run in the exact same parameterization in space---bridging prototyping and applications.

Big thank you to everyone who has been helping to bring TerraMind.tiny to orbit. When we started TerraMind over a year ago, I didn't expect it would end up circling Earth. 🚀

Dr. Juan Bernabe Moreno Thomas Brunschwiler Teodoro Laino Benedikt Blumenstiel Paolo Fraccaro

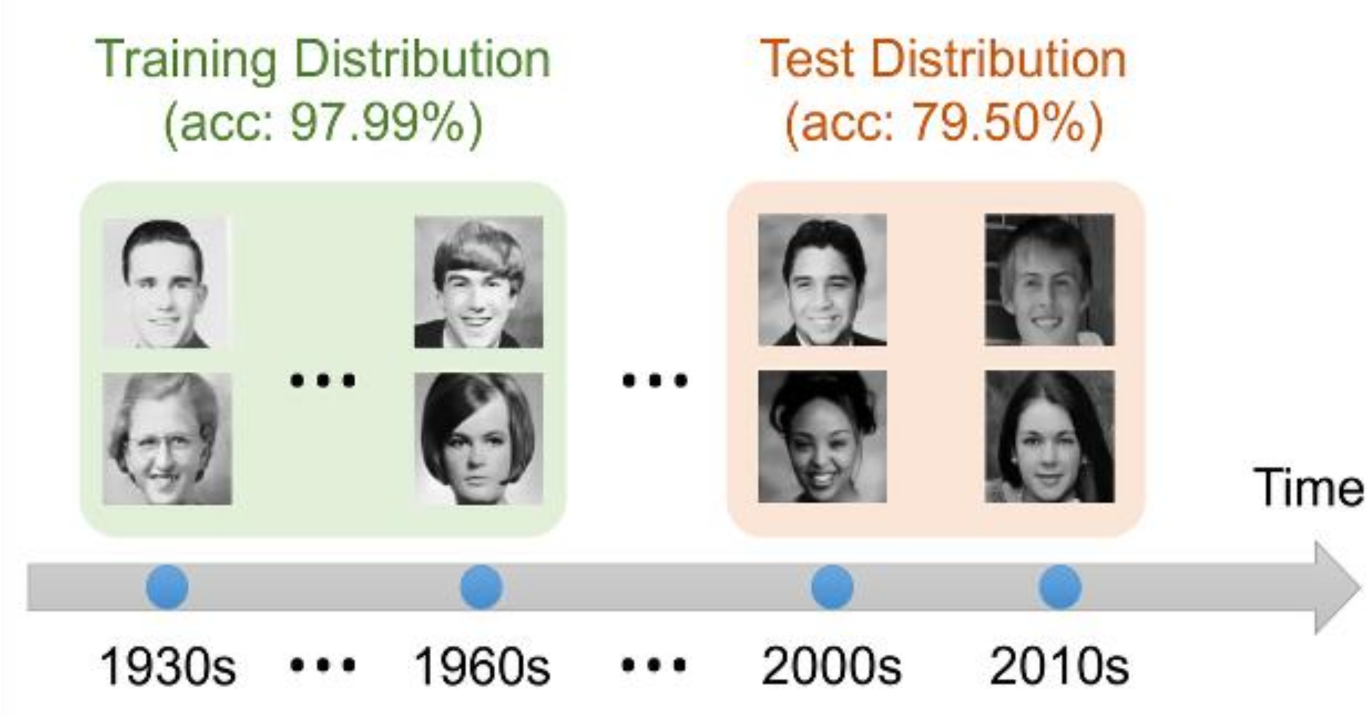
Links:
Tiny model annoucement: https://lnkd.in/epAcAm-f
Open-sourced models: https://lnkd.in/ei-M4iNw

Streaming Machine Learning for Multi-Modal Streams

Lorenzo Iovine, Giacomo Ziffer

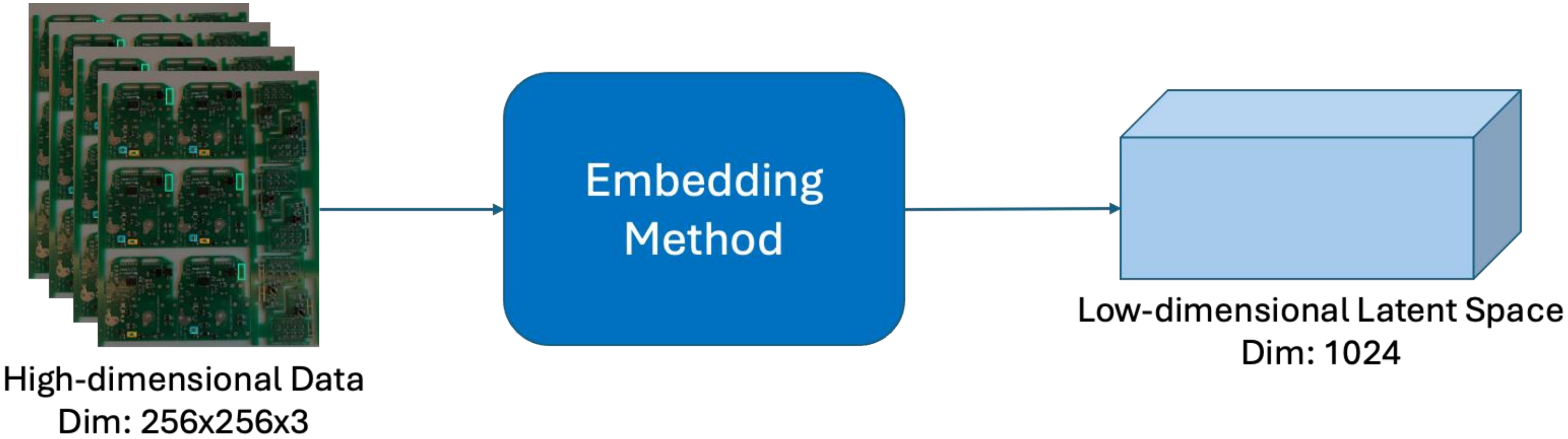
Challenge: temporal distribution shift

- The main challenge is that the data distribution changes over time: what a model learns today may not be valid tomorrow.



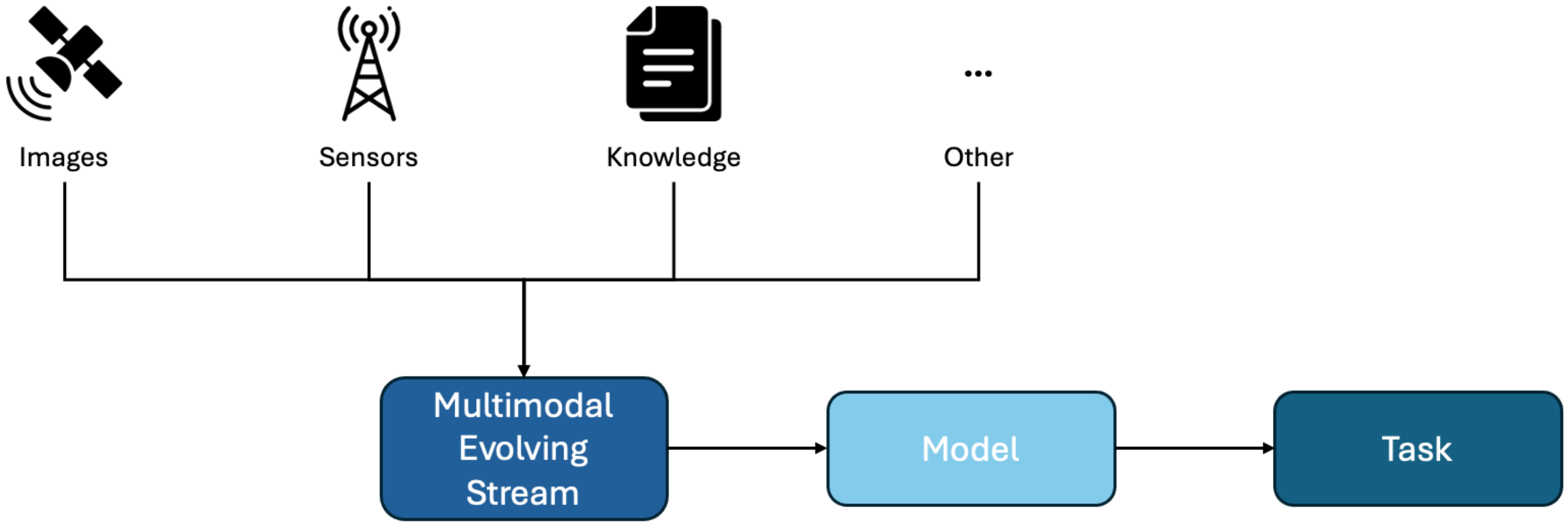
Challenge: high-dimensional data

- SML models must learn under strict time and memory constraints, they can struggle to operate on high-dimensional data, such as images.



Evolving Multi-Modal Streams

- In real applications, data streams are rarely unimodal. They often combine heterogeneous sources such as images, text, or metadata.

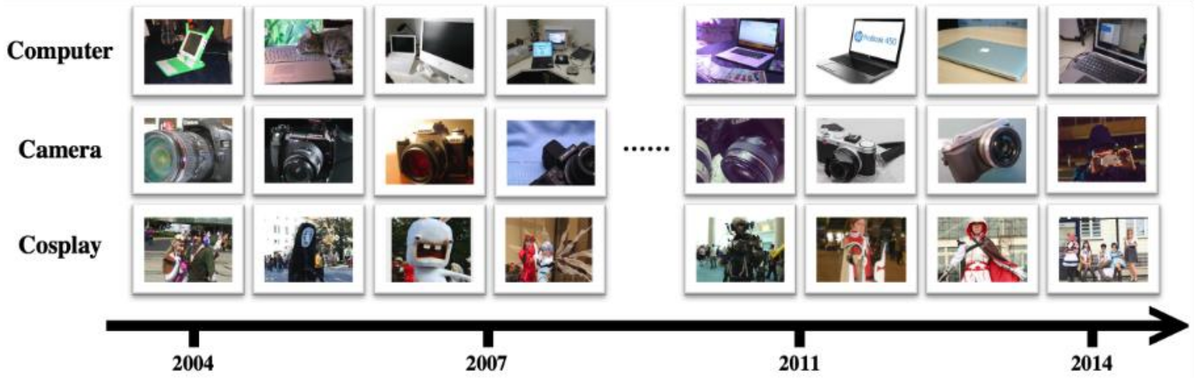


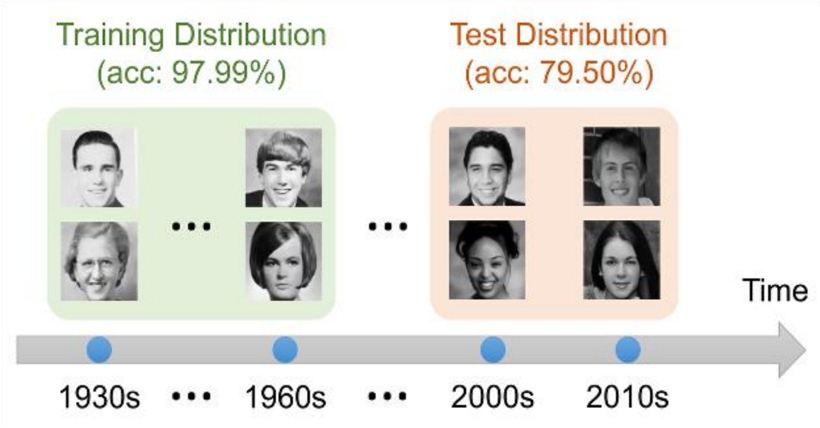
Research novel result thesis

Streaming Update of Unsupervised Image Embeddings

Goal

Study how unsupervised image embeddings can be updated in streaming at test time, while keeping the downstream classifier fixed and avoiding degradation of previously learned representations. The thesis focuses on a realistic streaming pipeline where images arrive continuously and their distribution changes over time.

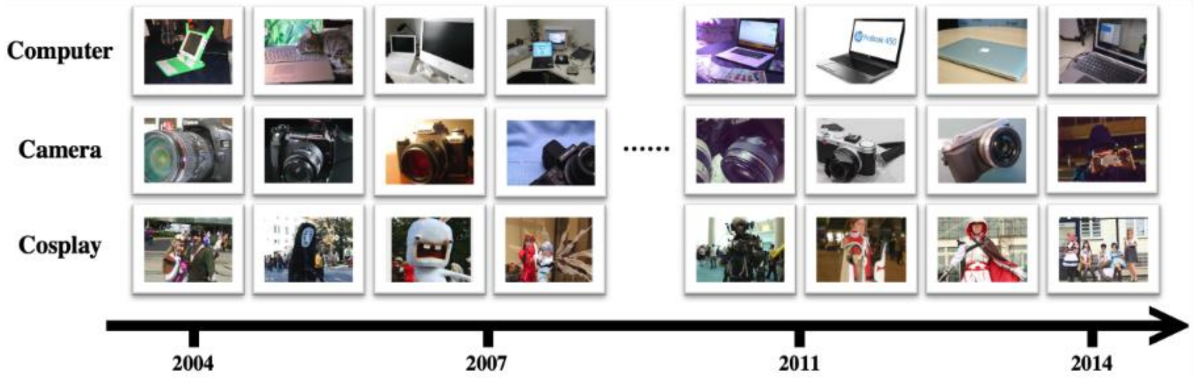


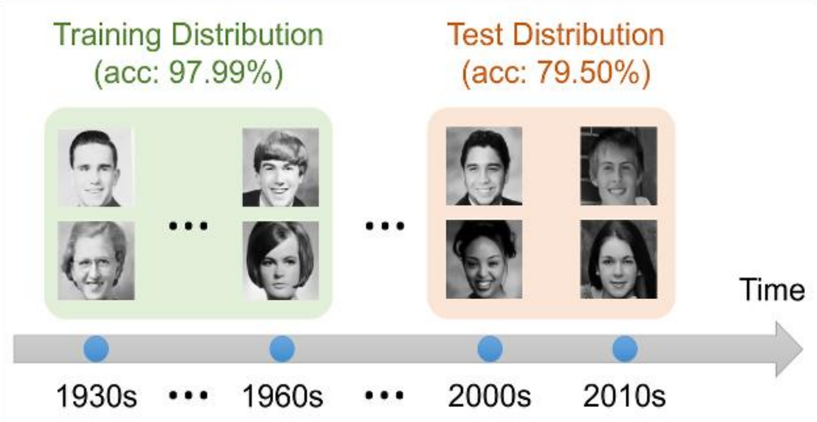


Research novel result thesis
Selective Adaptation in Streaming Image Classification

Goal

Design and evaluate an active online learning strategy for image streams, where model updates at test time are performed only on uncertain samples, while confidently predicted images are ignored to reduce labelling and training costs.





Research/Industrial novel result thesis

SML Regression for Global Time-Series Forecasting

Goal

The thesis investigates whether streaming regression models can effectively perform short-term forecasting on non-stationary time series (sales, stocks, indices, cryptocurrencies, …), where traditional batch forecasting methods often struggle with rapid distribution changes.

Motus ml’s Internship for Thesis

Giacomo Ziffer

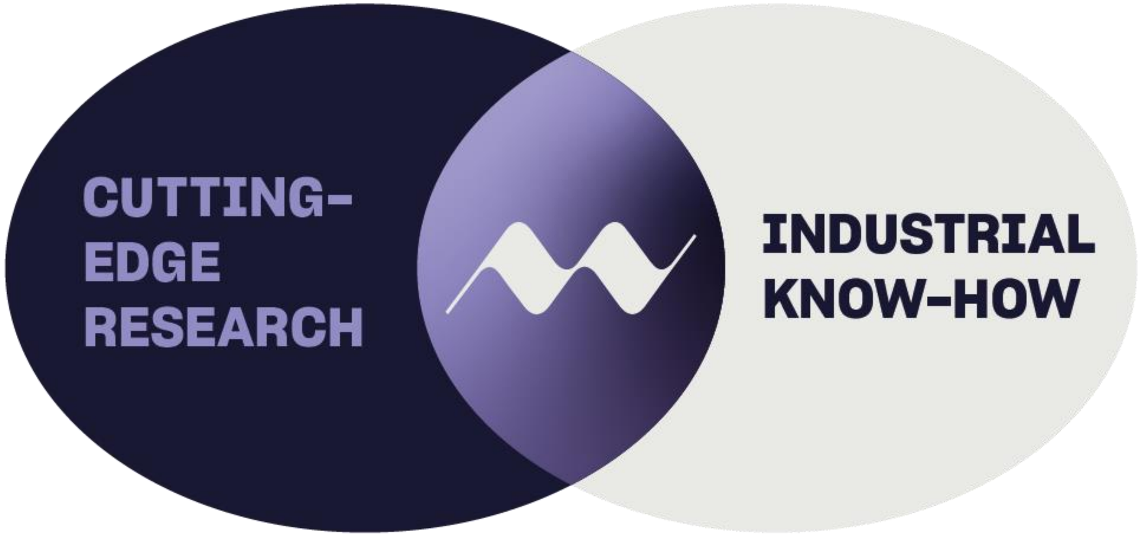
motus ml

The power of data, *at your fingertips*.

WWW.MOTUSML.COM

INFO@MOTUSML.COM

Who we are



- Research spin-off of **Politecnico di Milano**
- **Bridge** the industry with **academic research** through strong links with **university labs** and **joint industrial PhD** programs
- Developing AI algorithms that **continuously learn** and **adapt** to **dynamic environments**, enabling data stream analysis on **cloud platforms** and resource-constrained **edge devices**
- Enable **AI integration**, supporting processes from **design automation** to **smart subsystems** and **mission data intelligence**

On board AI-powered applications

Edge EO data analysis

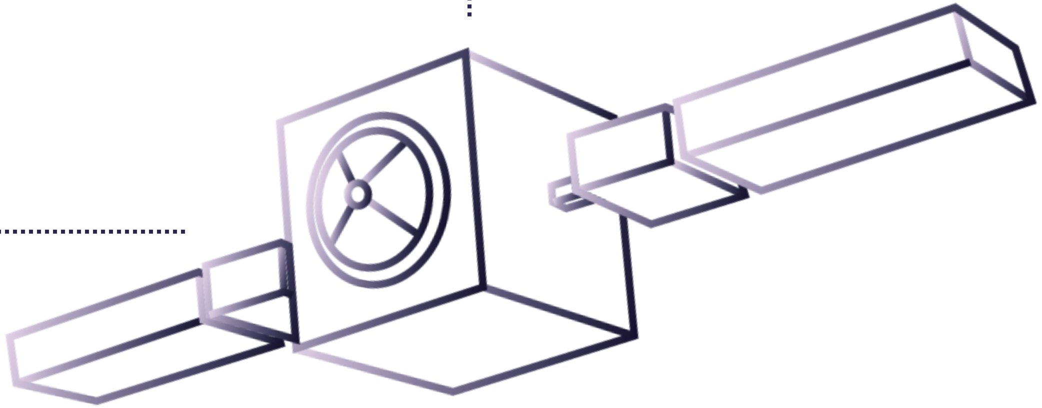
Solving Earth Observation tasks on board in **real-time**, maximizing value creation while **minimizing downlink cost** and **processing unit weight**.

Case: Cloud-Free Earth Observation

Situational awareness & autonomy

Fusing external **sensor** data with **telemetry** and **housekeeping** information to continuously analyze the environment, learn accurate prediction models, and determine actions to **prevent contingencies** affecting **satellite operability** and **safety**.

Case: Collision Avoidance



On ground analysis

Real-time Change Detection

Identifying and **dating** areas of **change** by continuously analyzing sequences of satellite images over time, **enabling rapid** disaster **response** and proactive environmental **monitoring**.

Case: Flood Monitoring and Prediction

Continuous Data Fusion

Incrementally processing **multi-source** and **multispectral** satellite imagery to extract key features, enabling **adaptive model** training and enhanced monitoring capabilities.

Case: Asbestos Roof Condition Monitoring

Adaptive Object Recognition & Tracking

Ensuring **reliable detection** and **tracking** of objects that **change shape** over time due to natural events or human-driven transformations.

Case: Construction Vehicles Monitoring

Smarter Signals for BepiColombo

Within the BepiColombo mission to Mercury, we utilize AI to analyze **flight telemetry** and **150+ health** monitoring **signals**, **reducing** by a factor of 10 the **thermoelastic noise** in the **ISA accelerometer** by a factor of 10.

We enable more **accurate and reliable** measurements of the **gravitational field of Mercury** for future scientific analysis.

motus ml AT A GLANCE

Developing AI algorithms that **continuously learn** and **adapt** to dynamic environments

Leveraging a streaming approach to **maximize performance** and **efficiency**

Prioritizing sustainable, **low-footprint AI solutions**

Internship Opportunities

Industrial novel result/use case thesis

Efficient Algorithms for Edge EO Data Analysis

Goal

The thesis investigates the development of high-efficiency algorithms for critical Earth Observation tasks, e.g., ship wake detection, specifically optimized for Edge Computing architectures.
The research addresses the challenge of deploying advanced ML approaches on resource-constrained systems, aiming to enable autonomous, real-time decision-making by bypassing the latency and bandwidth bottlenecks inherent in traditional ground-station data transmission.

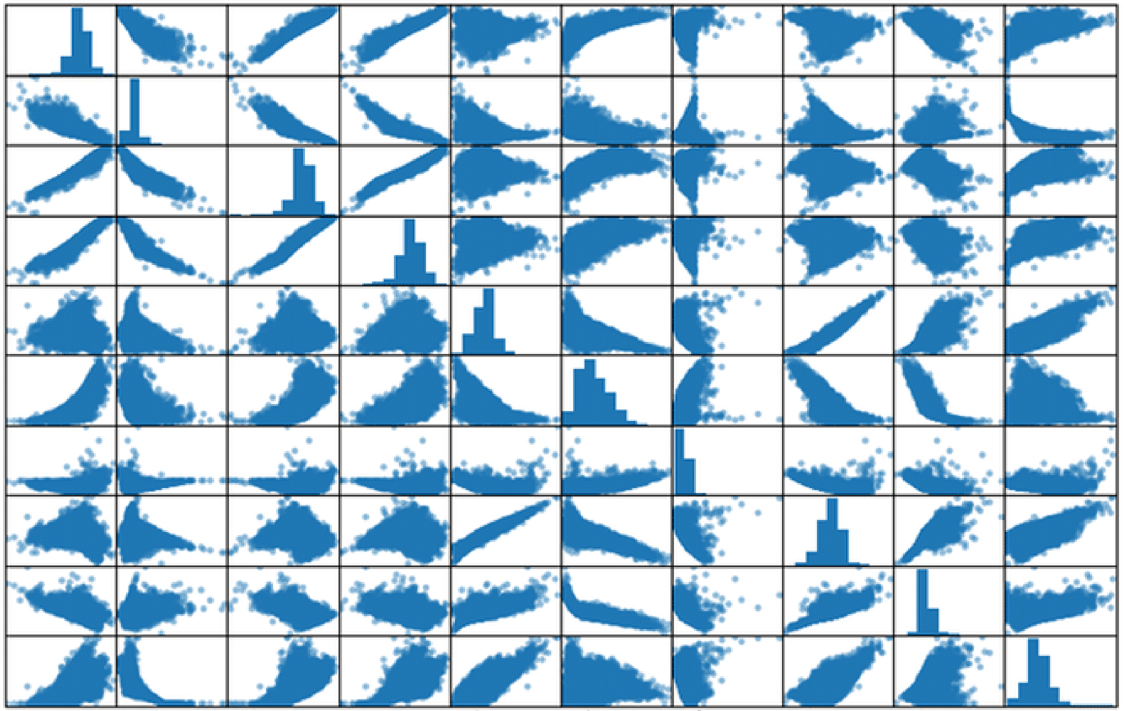
Industrial use case thesis

Time Series Forecasting of Spacecraft Housekeeping Telemetry

ThalesAlenia Space
a Thales / Leonardo company

Goal

Conducted in collaboration with **Thales Alenia Space Italia**, this thesis investigates advanced time series forecasting techniques applied to spacecraft housekeeping telemetry. The project aims to analyze historical data streams from onboard subsystems, such as thermal, power, and attitude control units, to predict future trends and potential drifts in satellite performance.



Streaming Data Engineering

Emanuele Della Valle

Research/Industrial theses

Riccardo Tommasini's offering in

Goal

Integrating stream processing and graph databases, e.g. Seraph [1] (joing work with Neo4j), to free users from manually specifying time windows [2].

[1] https://doi.org/10.48786/edbt.2024.21

[2] see slide deck 26b - prof-Tommasini-Thesis-in-Lion.pdf on webeep

No Erasmus needed, INSA Lyon provides the funding.

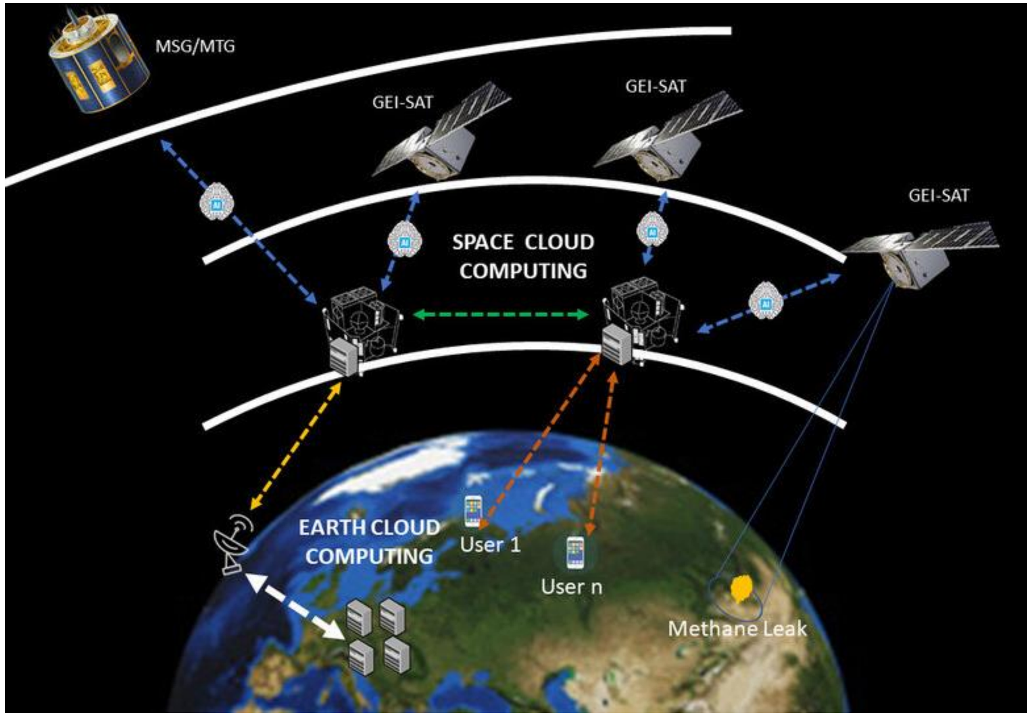
Contact: riccardo.tommasini@insa-lyon.fr

Research/Industrial theses

Scalable Stream Processing for Space Cloud

Goal

Rethinking topic-centric Event-Based Systems, DSMSs, CEPs, and Event-Driven Architectures for in-orbit cloud computing infrastructures under severe resource constraints (mass, volume, inter-satellite bandwidth, power, and thermal dissipation)



https://www.esa.int/Enabling_Support/Preparing_for_the_Future/Discovery_and_Preparation/ESA_extends_AI_and_cloud_computing_to_space

Streaming Data Analytics
Thesis proposal

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